12B2 — Deep-Dive Feature Audit: Skill/Combat VFX & Item Art Atlas Ecosystem (Dedicated Visual Supplement)

`ITEM_ART_CATALOG.json` (`9` WebP Atlases / `576` Cells), Semantic Icon-vs-Gameplay Contradictions, The `54`-Tech `[MOVE-CAT-01]` VFX Regression Trap, `2E/3E/4E` Fusion VFX Scaling, Domain Ground Rings & `24`-Slot Queue Priority

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Primary Audited Files**: `docs/ITEM_ART_CATALOG.json`, `tools/build_item_art.js`, `js/ui.js` (`resolveItemArt`, `renderAtlasIconHtml`), `js/skills.js` (`T1..T7_VFX`, `SUBPATH_VFX_MOTIFS`, `EQUIP_EFFECT_VFX_CATALOG`, `COMBAT_ACTION_VFX_CATALOG`), `js/render.js` (`updateCastEffects`, `drawCastEffects`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)
- **Scope**: Dedicated deep-dive into **Item Icons/Atlases** and **Skill/Equipment/Action VFX** (`zero audio`, `100%` read-only / zero-RNG render purity).

PART I — ITEM ART & SPRITE ATLAS AUDIT (`ITEM_ART_CATALOG.json` & `js/ui.js`)

1. `[ART-CAT-01]` Path Manual Collisions, Unused Cells & Missing Capstone Scrolls in `ITEM_ART_CATALOG.json`!

Inspecting all `576` cells across the `9` WebP sheets in `docs/ITEM_ART_CATALOG.json` exposes **four exact catalog mapping defects**:

1. **`GOD` & `DIVINE_SUMMONER` Collide on Cell `34` (`"summoner"`) While Cell `12` (`"alchemy"`) Is Orphaned (`lines 585–630`)**:
    - In `manualPaths`, both `"GOD": 34` (indigenous Chinese Incense/Shrine Deity path) and `"DIVINE_SUMMONER": 34` (Pantheon Summoner) point to cell `34` (`"summoner"`), even though `sheets.manuals` has unused thematic cells (`e.g., cell 61 "samsara-treasure" / cell 62`

"official-decree" / cell 60 "phoenix-treasure"), and **cell 12 ("alchemy") is referenced by 0 paths in manualPaths**!

2. **Missing Legendary Scroll Icons for LEG_V240_GOD & LEG_V240_CRAFT_CREATION (lines 662–713)**:
   - legendaryTechniques maps 50 legendary techniques (0..49 on sheets.legends), **omitting LEG_V240_GOD and LEG_V240_CRAFT_CREATION** (added in v2.4.0)! In inventory/library grids, 41 paths' capstone manuals display ornate Legendary scrolls while GOD and CRAFT_CREATION capstones downgrade to generic manuals!
   - **Test-Safe Fix**: Keep Object.keys(manifest.legendaryTechniques).length === 50 if asserted by legacy tests, and map LEG_V240_GOD (-> manuals cell 60 "phoenix-treasure" with grade: 'Legendary' gold finish) and LEG_V240_CRAFT_CREATION (-> manuals cell 47 "dragon-treasure" with grade: 'Legendary' finish) inside UI.resolveItemArt!
3. **SWORD Is the ONLY Equipment Family Without a Distinct 8th Mythic Cell (lines 716–727)**:
   - All 27 other equipment families in tierFamilies have 8 distinct cells ([t1..t7, mythic]), whereas "SWORD" has [56, 57, 58, 59, 60, 61, 62, 62] (duplicating cell 62 while "sword" cell 27 on sheets.equipment goes unused!).
4. **31 Grand Dao Divine Relics (REL_V230_<PATH>) Lack Relic Role & Dedicated Visual Identity**:
   - In UI.resolveItemArt, VC.LEGENDARY_EQUIPMENT (27 items) resolve role: 'relic' on sheets.equipment[0..26], while the 31 REL_V230_<PATH> Divine Relics fall back to shared generic tier art. Ensure all 31 REL_V230_* resolve to the **8th (-mythic) slot** of their weapon family with grade: 'Legendary' + path-colored seal badge!

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2. [ART-SEM-02] Semantic Icon-vs-Gameplay Contradictions (Player Sees One Weapon Type, Engine Computes Another!)

1. **The "พัดหยกสื่อปราณ" (FOCUS Drawn as FAN) & "กรงเล็บพยัคฆ์" (FIST Drawn as CLAW) Visual Trap**:
   - As documented in game-ui-ux, UI.resolveItemArt allows a FOCUS item with "พัด" in nameTh to display the **FAN (พัดยุทธ์) artwork**!
   - **Why This Hurts Players & AI Debugging**: Procedural FOCUS rolls "พัดหยกสื่อปราณ" (economy.js:586) with gearType: 'FOCUS'. The inventory shows a **Fan icon**, yet equipping it for a FAN martial art incurs the **0.65× wrong-weapon penalty** (04C)! The exact same visual contradiction occurs on procedural FIST items named "กรงเล็บพยัคฆ์" (icon vs CLAW skill penalty, 04B).

- **Exact Fix**: Rename the colliding `FOCUS` / `FIST` procedural `baseForms` in `economy.js:583–586` (`e.g. แส้ปัดขนจามรีเซียน / สนับหมัดพยัคฆ์คำราม`) AND resolve equipment icons strictly from **normalized `gearType`** (`never overriding` gearType: 'FOCUS'`with`'FAN' `based on flavour text!`).

2. **Paper Talismans (`ITEM190_TAL_001..024`) & Formation Cores (`ITEM190_CORE_001..012`) Drawn as Jewelry Charms**:
   - `sheets.materials` already has a painted **`"talisman"` cell (`cell 55`)**, yet `ITEM190_TAL_001..024` (`subType: 'CHARM'`) render as **jade belt-pendants (`supplies charm-t1..t7`)**! Route `ITEM190_TAL_*` (or `isTalismanConsumable`) to `"talisman"` (`with tier/grade frame`), and route `ITEM190_CORE_*` (`Formation Flags/Cores`) to `AUX_TOOL` (`or array disk`) so Talismans look like paper talismans, not necklaces!
3. **Unused Painted Elemental/Specialty Pill Cells (`equipment[56..61]` & `legends[61..63]`)**:
   - Route `ITEM190_PILL_012` (`Blood Overdrive`) $\to$ `equipment:56` (`pill-blood`), `EFX_LIFESPAN_EXTEND` (`Longevity`) $\to$ `equipment:58` (`pill-longevity`), `PILL_V235_FUSION_*` / `PILL_V234_*` elemental pills $\to$ `pill-ice` / `pill-fire` / `pill-lightning` / `pill-demonic` when matching element!
4. **Decoded Texture RAM Footprint (`10` Atlases $\approx 37.5\text{ MiB}$ RGBA on Mobile)**:
   - `4` `1254×1254` sheets (`~25.1 MiB` RGBA) + `5` `768×768` sheets (`~11.8 MiB` RGBA) + legacy `512×832` PNG (`~1.7 MiB` RGBA) = **`~38.6 MiB` uncompressed GPU/RAM texture memory** when all sheets decode simultaneously. Add `decoding="async"` / lazy CSS sheet mounting and avoid adding a 10th bitmap sheet (`use Canvas vector seals/badges instead`).

PART II — SKILL, EQUIPMENT & COMBAT ACTION VFX AUDIT (`js/skills.js` & `js/render.js`)

3. `[VFX-TRAP-03]` CRITICAL CROSS-PATCH REGRESSION TRAP: Fixing the `54` Mis-Tagged `ART190_*` Techniques (`03A`) Will Break `v260_vfx_tier_subpath_equip_suite.js` Unless Their VFX Maps Move Simultaneously!

- **Why GPT & Gemini MUST Read This Before Touching `data_tech.js` (`[MOVE-CAT-01]` in `03A`)**:

- In `JS2-v2.6.0`, `54` combat techniques (`every 4th entry in ART190_*`, e.g., เพลงทวนมังกรโลหิตกวาดทัพ) were accidentally tagged `QI_MANTRA` / `BODY_MANTRA` / `FAITH_MANTRA`, giving them `SK.profile(t) === null`.
- Consequently, `V2_6_0_COMPLETE_TIER_SUBPATH_EQUIP_VFX_REGISTRY` placed those `54` IDs into **`SK.PASSIVE_TO_T2..T6_VFX`** instead of **`SK.TECH_TO_T1..T7_VFX`**!
- Look at Assertion #2 in `test/v260_vfx_tier_subpath_equip_suite.js:48-59`:

```javascript
const ordinaryActive = JD.TECH.techniques.filter(t => !t.legendary &&
SK.profile(t));
for (const t of ordinaryActive) {
  // Asserts resolveCommonVfx / resolveT1..T7Vfx(t) is NON-NULL!
}
```

- **Mandatory Co-Update Rule**: In the exact same commit where those `54` `ART190_*` techniques are restored to `cat: 'COMBAT_TECHNIQUE'` (`making SK.profile(t) non-null`), **migrate all `54` IDs from `PASSIVE_TO_T*_VFX` into the matching weapon/element family in `SK.TECH_TO_T1..T7_VFX`** so `v260_vfx_tier_subpath_equip_suite.js` passes `100%` on the first run!

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4. `[VFX-REACT-04]` Flat Generic VFX on All `34` `2E/3E/4E` Elemental Fusion Reactions, `30` Shared Path Finishers & Invisible Battlefield Domains (`js/render.js`)!

1. **All `34` Elemental Fusion Reactions (`2E`, `3E`, `4E`) Share One Small Generic `'ELEMENTAL_REACTION_BURST'`**:
   - Triggering a 4-Element `4E-01 Genesis Creation` or `4E-03 Samsara Judgment` (`hardest combo in the game!`) plays the same generic burst as a 2-Element reaction!
   - **Fix**: Pass `reactionTier` (`2`, `3`, `4`), `reactionId`, and `elements[]` in `lastElementalReaction`, scaling radius (`2E: 28 px`, `3E: 42 px` trigram ring, `4E: 58 px` quad-pillar cosmos burst) and blending the exact `2-4` element colors from `ELEM`!
2. **`43` Path Gauge Finishers Crammed Into `13` `FINISHER_*` Forms**:
   - `COMBAT_ACTION_VFX_CATALOG` only defines `13` `FINISHER_*` shapes for `43` paths. In `drawCombatActionVfx` for `fin:npc:*` (`which already carries` fx.path`!`), layer **`drawPathApexCrownGeometry(ctx, Render.PATH_VISUAL_SIGNATURES[fx.path], ...)`** inside the finisher

burst so **all `43/43` paths display their exact iconic Grand Dao emblem on `100`-Gauge Finishers**!

3. **Invisible `f.activeDomain` Ground Boundaries (`8` Composable Battlefield Domains)**:
   - Render a restrained, low-alpha (`globalAlpha = 0.16`) runic ground perimeter circle for `f.activeDomain` (`skipped on` quality === 'low'`, static on` reducedMotion`) so players visually see the active Domain zone during fights!

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5. `[VFX-QUEUE-05]` `24`-Event `castActive` Queue Eviction Priority & Phantom Off-Turn DoT Conduit Beams (`js/render.js` & `js/combat.js`)!

1. **Priority-Aware `24`-Slot Queue Eviction in `Render.updateCastEffects`**:
   - `castActive` is hard-capped at `24` concurrent VFX (`verified in legendary_visual_identity_suite.js:76`). In large `8v8` raids, rapid `T1/T2` slashes and `iproc` item procs can fill all `24` slots—blocking a newly cast **Legendary Scene (`2,400 ms`), T7 Ultimate, Duet Resonance Strike, or Path Finisher**!
   - **Exact Fix**: Keep the strict `castActive.size <= 24` cap (`preserving all test assertions!`), but when `castActive.size === 24` and a high-priority event (`legendary || tier >= 6 || kind ∈ ['duet', 'finisher', 'reaction']`) arrives, **evict the oldest `tier <= 2` / ordinary `iproc` entry** to make room!
2. **Eliminate Phantom Cross-Map Mantra/Defense/Item VFX From `statusTick` DoT (`09C` Cross-Link)**:
   - Suppressing on-hit/reflect procs when `opts.isDot === true` (`09C`) stops characters from firing `MANTRA_VAJRA_BELL` / `DEF_TAIJI` / `conduit` beams across `25` tiles every minute poison ticks!